

Overview of Diabetes Mellitus:

Diabetes is a chronic metabolic condition that affects how the body turns food into energy. When blood sugar levels rise, the pancreas normally produces insulin to allow blood glucose into cells for use as energy. In diabetes, the body either does not make enough insulin or cannot use its own insulin effectively.

Primary Classifications:

1. Type 1 Diabetes: An autoimmune reaction where the body's immune system attacks insulin-producing beta cells in the pancreas. Requires daily exogenous insulin administration.
2. Type 2 Diabetes: The body cells become resistant to insulin, and the pancreas cannot produce enough insulin to overcome this resistance. Develops over years and is often preventable.
3. Gestational Diabetes: Develops during pregnancy in females who did not previously have diabetes.

Clinical Diagnostic Criteria:

- Fasting Plasma Glucose (FPG):
 - * Normal: Below 100 mg/dL
 - * Prediabetes: 100 mg/dL to 125 mg/dL
 - * Diabetes: 126 mg/dL or higher on two separate tests
- Hemoglobin A1C (Average Blood Sugar over 3 Months):
 - * Normal: Below 5.7%
 - * Prediabetes: 5.7% to 6.4%
 - * Diabetes: 6.5% or higher on two separate tests

Prediabetes and Prevention Strategies:

Prediabetes is a serious health condition where blood sugar levels are higher than normal, but not yet high enough to be diagnosed as type 2 diabetes. Approximately 96 million American adults - more than 1 in 3 - have prediabetes, and over 80% do not know they have it.

CDC National Diabetes Prevention Program (National DPP):

Clinical trials demonstrate that people with prediabetes can cut their risk of developing type 2 diabetes by 58% (71% for adults aged 60 and older) by making modest lifestyle changes:

- Weight Reduction: Losing 5% to 7% of starting body weight (about 10 to 14 pounds for a 200-pound person).
- Physical Activity: Engaging in at least 150 minutes per week of brisk walking or comparable moderate physical activity.
- Healthy Nutrition: Adopting a diet rich in non-starchy vegetables, lean proteins, whole grains, and water instead of sugar-sweetened beverages.

Modifiable Risk Factors for Type 2 Diabetes:

- Overweight or obesity
- Being physically active less than 3 times a week
- High intake of ultra-processed carbohydrates and sugary drinks
- History of gestational diabetes or polycystic ovary syndrome (PCOS)